PhenoBridge: Confidence-Aware Random-Walk Fingerprints on AlphaFold Contact Graphs for Genotype–Phenotype Analysis in KBG Syndrome

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Abstract. KBG syndrome is a rare autosomal-dominant neurodevelopmental disorder caused by heterozygous variants in *ANKRD11*, a 2,663-residue protein that is mostly intrinsically disordered. Missense and truncating variants are spread across the gene, and the clinical picture varies widely between patients. Whether a variant's structural context explains part of that variation is still an open question. We present PhenoBridge, a lightweight and interpretable pipeline that turns each variant position into a *structural influence fingerprint*. It builds a Cα contact graph from the AlphaFold model of ANKRD11, partitions residues by pLDDT confidence, and runs a random walk with restart (RWR) from the variant residue. The fingerprint records how much of the walk's stationary probability falls in high-confidence core, intermediate-confidence, disordered and inter-domain "backbone" residues. We applied PhenoBridge to a curated cohort of 333 individuals annotated with 48 Human Phenotype Ontology (HPO) terms. Fingerprints change very little when the restart probability or contact cutoff is varied (Spearman $\rho \geq 0.986$). However, 84% of cohort variants sit in low-confidence regions (pLDDT < 50). In per-phenotype logistic models adjusted for variant class, no HPO term was associated with the fingerprint after false-discovery-rate correction. The two nominal signals appeared only when unannotated phenotypes were coded as absent, and they disappeared when the analysis was restricted to explicitly recorded observations. We release the pipeline as an interactive tool and discuss what structure-only features can and cannot offer for genotype–phenotype work in proteins dominated by disorder.

Keywords: KBG syndrome; ANKRD11; AlphaFold; pLDDT; residue contact networks; random walk with restart; Human Phenotype Ontology; genotype–phenotype correlation.

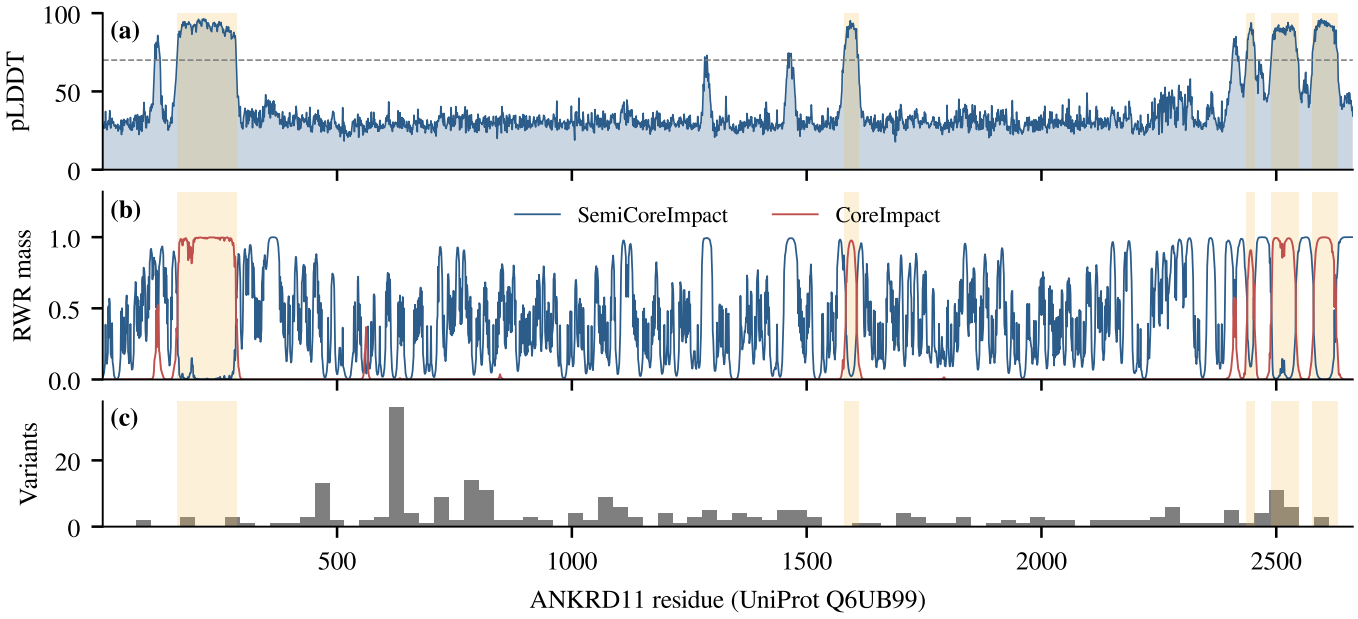


Figure 1. Structural landscape of ANKRD11 and the cohort variants. (a) AlphaFold pLDDT along the sequence; the dashed line marks pLDDT = 70. (b) RWR fingerprint components F_{Semi} (blue) and F_{Core} (red) computed with every residue in turn as the seed ($\alpha = 0.3$, $r = 8$ Å). (c) Positions of the 256 cohort sequence variants that could be mapped to a residue. Shaded bands mark confident segments (pLDDT ≥ 70, ≥20 residues): 159–285, 1579–1610, 2436–2455, 2489–2547 and 2577–2631.

1 Introduction

KBG syndrome (OMIM 148050) is characterised by macrodontia of the upper central incisors, distinctive craniofacial features, short stature, skeletal anomalies and developmental delay or intellectual disability. It is caused by heterozygous loss-of-function variants in *ANKRD11* or by 16q24.3 deletions that include the gene [1,2]. ANKRD11 is a large chromatin regulator. Apart from an N-terminal ankyrin-repeat domain and short C-terminal segments, it has few well-folded regions. Large cohort studies have described broad clinical variability and only modest genotype–phenotype correlations [3].

AlphaFold [4] and the AlphaFold Protein Structure Database [5] now provide a predicted structure for essentially every human pro-

tein, together with a per-residue confidence score (pLDDT). pLDDT tracks local order closely: residues below about 50 are a strong indicator of intrinsic disorder [6,7]. This makes it possible to place every patient variant in structural context, even for proteins that have no experimental structure. The question we address is practical. Can a simple, transparent structural feature computed from an AlphaFold model separate patients by phenotype within a single-gene cohort?

We borrow ideas from two areas. Residue contact networks built from Cα coordinates underpin elastic-network models of protein dynamics [8], and random walks with restart (RWR) are a standard way to propagate a local signal through a network [9,10]. PhenoBridge combines the two. We seed a walk at the variant residue on the contact graph and summarise where the probability mass settles, measured against regions defined by pLDDT. Our contributions are: (i) an interpretable four-component fingerprint for any residue; (ii) an open, end-to-end pipeline from a phenopacket-style cohort file to per-HPO association statistics and an interactive prediction interface; and (iii) a candid evaluation on 333 KBG individuals, including how sensitive the result is to the coding of missing phenotype data.

2 Data

Structure. We used the AlphaFold DB model AF-Q6UB99-F1 (v6) for human ANKRD11 (UniProt Q6UB99, 2,663 residues). AlphaFold stores per-residue pLDDT in the B-factor column. PDB numbering matches UniProt numbering, so cohort positions map directly onto residues.

Cohort. The cohort file (`ANKRD11_KBGS_individuals.json`) contains 333 individuals with a KBG syndrome diagnosis, curated from the literature, primarily the 340-patient series of Martinez-Cayuelas *et al.* [3]. Each individual is annotated against 48 HPO terms [11] as *observed*, *excluded* or *not available*, and carries one *ANKRD11* allele (transcript NM_013275.6). There are 232 distinct alleles: 77 individuals carry structural variants (16q24.3 deletions), and the remainder carry sequence variants given as cDNA HGVS. Phenotype annotation is sparse: the median HPO term is unrecorded for 70.7% of individuals (maximum 99.1%).

3 Methods

3.1 Contact graph and confidence regions

Let x_i be the Cα coordinate of residue i. The contact graph $G = (V, E)$ has one node per residue and an undirected edge whenever $\|x_i - x_j\| \le r$, with $r = 8$ Å by default. Edges are found with a k-d tree. Sequence neighbours (≈3.8 Å apart) are always connected, so G contains the chain as a path plus the spatial contacts. Residues are split by the empirical pLDDT quantiles $q_{0.9}$ and $q_{0.5}$: *Core* (pLDDT ≥ $q_{0.9}$), *Semi-core* ($q_{0.5}$ ≤ pLDDT < $q_{0.9}$) and *Disorder* (pLDDT ≤ $q_{0.5}$). We use quantiles rather than fixed thresholds because they adapt to proteins of different overall confidence.

3.2 Communication backbone

To mark residues that may relay signals between distant domains, we build a second graph G_{core} restricted to residues with pLDDT ≥ 70. We then sample up to 2,000 source–target pairs between an N-terminal anchor (residues 1–200) and a C-terminal anchor (the last 200 residues). For each pair we compute the shortest path and count how often each interior node is visited. The 200 most frequently visited nodes form the *communication backbone B*.

3.3 RWR fingerprint

Let A be the adjacency matrix of G and D its degree matrix. For a variant at residue s, let $\boldsymbol{e}_s$ be the indicator vector of s. We iterate

$$\boldsymbol{p}^{(t+1)} = (1 - \alpha)\, A\, D^{-1}\, \boldsymbol{p}^{(t)} + \alpha\, \boldsymbol{e}_s \tag{1}$$

until the change in ℓ_1 norm falls below 10^{-10} (or 200 iterations), with restart probability $\alpha = 0.3$. The fixed point $\boldsymbol{p}^{\infty}$ is a personalised-PageRank profile centred on s [9]. For each region $R \in$ {Core, Semi-core, Disorder, B} the fingerprint component is

$$F_R(s) = \Sigma_{i \in R}\, p_i^{\infty} \tag{2}$$

The three pLDDT regions partition V, so their components sum to one, and F_B overlaps them. Because the walk restarts at the seed, the fingerprint describes the variant's local neighbourhood within a radius set by α. F_{Semi} is high for residues embedded in partially ordered structure and low both for residues in long disordered linkers and for residues deep inside the confident core.

3.4 Variant mapping and association model

Protein positions are read from `p.` notation when it is present. Otherwise they are computed as $\lceil c/3 \rceil$ from the first cDNA coordinate. Alleles whose HGVS string contains `del`, `dup`, `ins`, `fs`, `ter/*` or `splice` are flagged as truncating-like ($T \in \{0,1\}$). For each individual we take the maximum F_{Semi} across their alleles. For every HPO term h whose observed and excluded counts were each at least 15, we fit

$$\text{logit}\, \Pr(y_h = 1) = \beta_0 + \beta_1 \log_{10} F_{\text{Semi}} + \beta_2\, T \tag{3}$$

and report the odds ratio $\exp(\beta_1)$ per 10-fold increase in F_{Semi}, Wald 95% confidence intervals, and Benjamini–Hochberg q-values [12]. We compare two ways of coding phenotypes. In **Encoding A**, used by our hackathon prototype, unannotated terms are treated as absent. In **Encoding B**, each model uses only individuals for whom the term was explicitly recorded as observed or excluded. Structural variants have no single residue position and are excluded from the association analysis.

3.5 Implementation

PhenoBridge is a single Python module built on Biopython [13], NetworkX [14], SciPy and statsmodels [15]. A Gradio interface exposes three functions: computing the fingerprint for a residue, comparing two residues side by side, and producing a top-10 table of predicted phenotype probabilities for user-entered variants, with a badge on terms whose cohort-level $p < 0.05$. A fingerprint takes about 0.06 s per residue on one CPU core, or about 2.5 min for the whole protein.

4 Results

4.1 ANKRD11 is predicted to be mostly disordered

pLDDT has a median of 31.2 and a mean of 39.2. 83.8% of residues fall below 50, and only 12.3% reach 70 or above (Fig. 1a). The confident residues cluster in an N-terminal segment (159–285), consistent with the annotated ankyrin-repeat domain, and in a short internal segment and three C-terminal segments. With the quantile thresholds ($q_{0.9}$ = 81.2, $q_{0.5}$ = 31.2), there are 268 Core, 1,066

Semi-core and 1,332 Disorder residues. The full contact graph has 2,663 nodes and 6,911 edges. The high-confidence graph G_{core} (327 nodes, 1,354 edges) splits into 10 connected components, so the backbone B (200 nodes, spanning residues 112–2442) runs through the largest connected confident regions rather than forming one continuous path from N- to C-terminus.

Table 1. Summary of structure, cohort and fingerprint statistics.

Quantity	Value
Residues / contact edges (r = 8 Å)	2,663 / 6,911
Residues with pLDDT ≥ 70 / < 50	12.3% / 83.8%
Core / Semi-core / Disorder residues	268 / 1,066 / 1,332
Individuals / HPO terms / distinct alleles	333 / 48 / 232
Structural variants (no residue position)	77
Individuals mapped / distinct positions	256 / 153
Truncating-like among mapped	69.5%
Median HPO missingness	70.7%
Seeds with pLDDT < 50 / ≥ 70	84.0% / 31 indiv.
F_{Semi} median (IQR), cohort	0.29 (0.16–0.57)
Spearman ρ vs. default: α = 0.15 / 0.5	0.988 / 0.998
Spearman ρ vs. default: r = 6 / 10 Å	0.986 / 0.992

4.2 Fingerprints are stable and interpretable

Fig. 1b shows that F_{Semi} varies sharply along the sequence. It is close to zero in long disordered stretches and close to one in the C-terminal third, where moderate-confidence regions are dense. F_{Core} is concentrated almost entirely in the confident segments. Fig. 2a compares the two default residues of the interactive tool. Residue 600 (pLDDT 27.5) keeps 94.7% of its walk mass in disorder (F_{Semi} = 0.053). Residue 2400 (pLDDT 49.4) sends 97.2% to the semi-core and 7.7% to the backbone. Across the 153 cohort positions, the ranking of F_{Semi} barely changes when α is halved or raised to 0.5, or when the contact cutoff is set to 6 or 10 Å (ρ ≥ 0.986; Table 1). The fingerprint is therefore a stable property of the structure, not an artefact of the parameter choices.

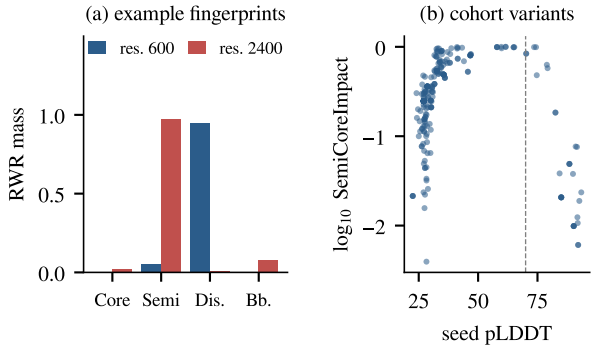


Figure 2. (a) Fingerprints for residue 600 (disordered linker) and residue 2400 (C-terminal semi-ordered region). Bb. = backbone. (b) $\log_{10} F_{\text{Semi}}$ against seed pLDDT for the cohort variants. Seeds inside confident domains (right) lose mass to the Core, so F_{Semi} is not a monotone function of pLDDT.

Most cohort variants are poorly resolved structurally. 84.0% of mapped individuals carry a variant at a residue with pLDDT < 50, and only 31 carry one in a segment with pLDDT ≥ 70 (Fig. 2b). Variants cluster around residues 530–670, 800–930 and 2400–2530 (Fig. 1c). Only the last of these clusters lies in a region with high F_{Semi}.

4.3 Phenotype association

Under Encoding A, 31 HPO terms met the prevalence filter. Two reached nominal significance: attention-deficit/hyperactivity disorder (OR 0.53, 95% CI 0.31–0.93, p = 0.026) and triangular face (OR 0.59, 0.35–0.99, p = 0.048). Neither survived FDR correction (minimum q = 0.60). Under Encoding B, the number of individuals per model fell from 256 to between 41 and 214. None of the 30 tested terms reached nominal significance. The smallest p-value was 0.15 (hypertelorism, OR 1.73, 0.82–3.65) and the minimum q was 0.83 (Table 2, Fig. 3). For intellectual disability, the core neurodevelopmental feature, the point estimate was above one (OR 1.61, 0.77–3.35), but the interval was wide and included one.

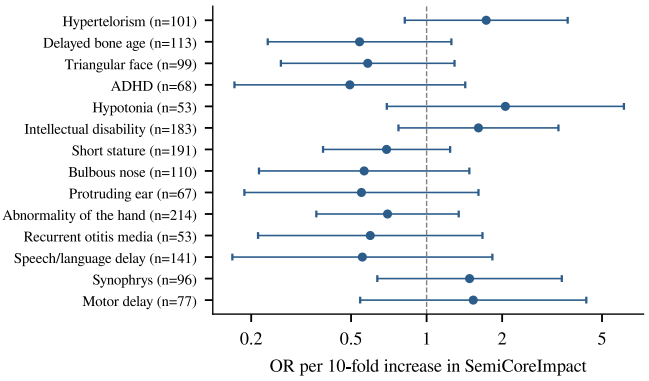


Figure 3. Odds ratios (95% CI) for F_{Semi} from Eq. (3) under Encoding B, for the 14 terms with the smallest p-values. n = number of individuals with an explicit annotation. Every interval includes OR = 1.

Table 2. Effect of phenotype coding on the strongest associations (OR per 10-fold increase in F_{Semi}, adjusted for T).

HPO term	Enc. A (n=256)		Enc. B (explicit only)		
	OR	p	n	OR	p
ADHD	0.53	0.026	68	0.49	0.19
Triangular face	0.59	0.048	99	0.58	0.18
Intellectual disability	0.96	0.86	183	1.61	0.20
Hypertelorism	1.31	0.36	101	1.73	0.15
Terms tested / p < 0.05 / min q	31 / 2 / 0.60			30 / 0 / 0.83	

5 Discussion

PhenoBridge yields a structural descriptor that is reproducible, cheap to compute and easy to read. In this cohort, however, that descriptor does not explain phenotypic variability among KBG patients. We see three reasons, and each points to a direction for future work.

Structural signal is limited. Most pathogenic *ANKRD11* variants are truncating and lie in disordered regions. For these variants, haploinsufficiency through nonsense-mediated decay probably matters more than local 3D context, and a contact graph built from low-confidence coordinates mostly encodes chain connectivity. Fingerprints are likely to be more informative for missense variants in the confident C-terminal segments. Testing that subset needs a larger cohort than the 31 individuals available here.

Missing data shape the conclusions. Treating unrecorded terms as absent (Encoding A) produced nominal hits that disappeared

when only explicit annotations were used. Literature-curated cohorts typically report a phenotype only when it was examined, so absence of a record is not evidence of absence. We recommend Encoding B, or an explicit missing-data model, as the default for this kind of analysis.

Model choices. Our variant-class flag is a string heuristic. It labels in-frame indels as truncating and misses single-nucleotide nonsense changes. A proper consequence annotation, for example from VEP, would remove this confound. Taking the maximum over alleles per individual is trivial here because each individual has one allele, but it would need revisiting for compound or mosaic cases. Finally, the per-term models are small and the 30 tests are correlated. Multivariate approaches such as phenotype-similarity scores or penalised multi-label models may recover a weak shared signal that individual tests cannot.

Outlook. The fingerprint does not depend on the gene. Applying it to rare-disease genes whose structures are better resolved, combining it with conservation and missense-pathogenicity scores, and weighting edges by predicted aligned error (PAE) instead of thresholded Cα distances are natural next steps.

6 Conclusion

We introduced PhenoBridge, an interpretable pipeline that links AlphaFold-derived structural context to patient phenotypes through confidence-aware random-walk fingerprints. The fingerprints are robust to parameter choices. In a 333-patient KBG syndrome cohort they show no phenotype association that survives multiple-testing correction. This negative result is informative: it shows that most *ANKRD11* variants fall in disorder, and that phenotype missingness can create spurious associations. Code, data and an interactive interface are available in the project repository.

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